

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: *Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)*

Assessment Tool Weightage: 40%

Annexure A

SHORT ANSWER TEST

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Code of Conduct:

I NS BHAVESH(192565086) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

NS BHAVESH

Signature of the student:

Annexure A

Short Answer Test

1. A company needs to transfer a **150 MB** design file from a client computer to a remote server over a network with a bandwidth of **50 Mbps**. The Transport Layer divides the file into **1500-byte segments**, while the Data Link Layer encapsulates each segment into frames by adding **40 bytes of header and trailer**. Calculate the total number of segments and frames transmitted, the total transmission overhead introduced by the Data Link Layer, and estimate the total transmission time (ignoring propagation delay). Explain how the Transport and Data Link layers ensure reliable communication.
2. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.
3. A **12,000-byte** IP packet is transmitted through a router where the Maximum Transmission Unit (MTU) is **1500 bytes**. Assuming each fragment requires a **20-byte IP header**, calculate the total number of fragments created and the total header overhead introduced. Explain how the Network Layer ensures successful delivery of fragmented packets to the destination.
4. A Transport Layer protocol uses a **sliding window size of 6 frames**. Each frame carries **1024 bytes** of user data. If the sender transmits **48 frames**, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

5. A multimedia application transfers a **600 MB** video file. The Session Layer inserts a synchronization checkpoint after every **60 MB** of transmitted data. During transmission, a failure occurs after **390 MB** has been transferred. Determine the number of checkpoints created before failure and calculate the amount of data that must be retransmitted after recovery. Explain the importance of synchronization in the Session Layer.

6. Before transmission, the Presentation Layer compresses a **900 MB** file by **40%**. The compressed data is encrypted, adding **5%** overhead to the compressed size. Calculate the compressed file size, the encrypted file size, and the total percentage reduction compared to the original file. Explain how the Presentation Layer improves transmission efficiency and security.

7. An FTP application transfers **3 GB** of data over a network with an effective throughput of **120 Mbps**. The Application Layer divides the data into requests of **3 MB** each. Calculate the total number of requests generated and estimate the total transmission time in seconds. Discuss how the Application Layer provides reliable services to users.

8. A packet experiences the following processing delays at different OSI layers: Application Layer = **4 ms**, Transport Layer = **3 ms**, Network Layer = **5 ms**, Data Link Layer = **2 ms**, and Physical Layer = **1 ms**. The propagation delay is **18 ms**, and the transmission delay is **12 ms**. Calculate the total end-to-end delay experienced by the packet and explain how each layer contributes to the communication process.

9. A network transfers **80 MB** of useful data using frames of **1600 bytes**, where each frame contains **80 bytes** of protocol overhead added by different OSI layers. Calculate the total number of frames required, the total overhead transmitted, and

the transmission efficiency as a percentage. Explain how protocol overhead affects network performance.

10.A hospital transfers a patient's medical record of **240 MB** through an IoT-enabled healthcare network. The Presentation Layer compresses the data by **30%**, the Transport Layer divides the compressed data into **1400-byte segments**, and the Data Link Layer adds **60 bytes** of overhead to each frame. Calculate the compressed file size, the total number of segments transmitted, the total protocol overhead introduced, and the final amount of data transmitted across the Physical Layer. Explain how multiple OSI layers work together to ensure secure and reliable communication.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates a thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers

Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand

Question 1

A company needs to transfer a 150 MB design file from a client computer to a remote server over a network with a bandwidth of 50 Mbps. The Transport Layer divides the file into 1500-byte segments, while the Data Link Layer encapsulates each segment into frames by adding 40 bytes of header and trailer. Calculate the total number of segments and frames transmitted, the total transmission overhead introduced by the Data Link Layer, and estimate the total transmission time (ignoring propagation delay). Explain how the Transport and Data Link layers ensure reliable communication.

Calculation:

File size = 150 MB = 150,000,000 bytes

Number of segments = $150,000,000 \div 1500 = 100,000$ segments

Number of frames = 100,000 (one frame per segment)

DLL overhead per frame = 40 bytes $\Rightarrow$ Total overhead = $100,000 \times 40 = 4,000,000$ bytes = 4 MB

Total data transmitted = $150,000,000 + 4,000,000 = 154,000,000$ bytes

In bits = $154,000,000 \times 8 = 1,232,000,000$ bits

Transmission time = $1,232,000,000 \div 50,000,000 \text{ bps} = 24.64$ seconds

Result:

- Total segments = 100,000
- Total frames = 100,000
- Total Data Link overhead = 4,000,000 bytes (4 MB)
- Total transmission time ≈ 24.64 seconds

Explanation:

The Transport Layer ensures reliable communication by breaking the file into fixed-size segments, numbering them sequentially, and using acknowledgements and retransmission (as in TCP) so that any lost or corrupted segment can be detected and resent, guaranteeing that the receiver reassembles the file correctly and in order.

The Data Link Layer ensures reliability over each physical hop by wrapping every segment in a frame with header and trailer fields that carry addressing and error-detection information (such as a CRC/checksum). This allows the receiving node to detect transmission errors introduced by the physical medium and request retransmission or discard corrupted frames, while also handling framing and flow control on the link.

Question 2

A Transport Layer protocol uses a sliding window size of 6 frames. Each frame carries 1024 bytes of user data. If the sender transmits 48 frames, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

Calculation:

Window size = 6 frames

Total frames to send = 48

Number of windows = $48 \div 6 = 8$ windows

Lost frames = 1 per window $\times 8$ windows = 8 retransmissions

Result:

- Number of transmission windows required = 8
- Total number of retransmissions = 8

Explanation:

Flow control (implemented through the sliding window mechanism) allows the sender to transmit only as many frames as the receiver can currently buffer and process, rather than flooding the link. This prevents buffer overflow at the receiver, reduces unnecessary packet loss, and keeps the sender and receiver synchronized in speed.

By allowing several frames (a whole window) to be sent before waiting for individual acknowledgements, flow control also improves throughput compared to stop-and-wait transmission, since the link stays utilized while earlier frames are still being acknowledged, improving overall communication efficiency.

Question 3

A 12,000-byte IP packet is transmitted through a router where the Maximum Transmission Unit (MTU) is 1500 bytes. Assuming each fragment requires a 20-byte IP header, calculate the total number of fragments created and the total header overhead introduced. Explain how the Network Layer ensures successful delivery of fragmented packets to the destination.

Calculation:

MTU = 1500 bytes

Header per fragment = 20 bytes

Usable data per fragment = $1500 - 20 = 1480$ bytes

Number of fragments = $\lceil 12,000 \div 1480 \rceil = \lceil 8.11 \rceil = 9$ fragments

Total header overhead = $9 \times 20 = 180$ bytes

Result:

- Total number of fragments = 9
- Total header overhead = 180 bytes

Explanation:

The Network Layer (IP) ensures successful delivery of fragmented packets by assigning each fragment the same identification number as the original packet, along with a fragment offset field that records its position within the original data stream, so the destination can reassemble the fragments in the correct order.

A More Fragments (MF) flag marks all but the last fragment so the receiver knows when reassembly is complete, and if any fragment is lost the entire original packet is discarded and can be requested again by upper-layer error recovery, ensuring end-to-end correctness of the delivered data.

Question 4

A Transport Layer protocol uses a sliding window size of 6 frames. Each frame carries 1024 bytes of user data. If the sender transmits 48 frames, calculate the number of transmission windows required. If one frame in every window is lost and must be retransmitted, determine the total number of retransmissions. Explain how flow control improves communication efficiency.

Calculation:

Window size = 6 frames

Total frames to send = 48

Number of windows = $48 \div 6 = 8$ windows

Lost frames = 1 per window $\times$ 8 windows = 8 retransmissions

Result:

- Number of transmission windows required = 8
- Total number of retransmissions = 8

Explanation:

As in Question 2, the sliding window mechanism regulates how much data the sender can push onto the network before receiving acknowledgement, matching the sender's rate to the receiver's processing capacity.

This flow control avoids overwhelming the receiver's buffer and the network path, minimizes congestion-related losses, and permits pipelining of frames, which together increase effective throughput and communication efficiency compared to sending and waiting for one frame at a time.

Question 5

A multimedia application transfers a 600 MB video file. The Session Layer inserts a synchronization checkpoint after every 60 MB of transmitted data. During transmission, a failure occurs after 390 MB has been transferred. Determine the number of checkpoints created before failure and calculate the amount of data that must be retransmitted after recovery. Explain the importance of synchronization in the Session Layer.

Calculation:

Checkpoint interval = 60 MB

Data transferred before failure = 390 MB

Checkpoints reached = $\lfloor 390 \div 60 \rfloor = 6$ (at 60, 120, 180, 240, 300, 360 MB)

Data since last checkpoint (360 MB) = $390 - 360 = 30$ MB

Result:

- Number of checkpoints created before failure = 6
- Data to be retransmitted after recovery = 30 MB

Explanation:

The Session Layer is responsible for establishing, managing, and synchronizing communication sessions between applications. By inserting periodic synchronization checkpoints, it allows the session to be resumed from the last confirmed checkpoint instead of restarting the entire transfer after a failure.

This dramatically reduces the amount of data that must be resent (only 30 MB instead of the full 600 MB in this case), saving bandwidth and time, and it also helps maintain session state such as dialogue control and recovery points during long-running transfers like video or file uploads.

Question 6

Before transmission, the Presentation Layer compresses a 900 MB file by 40%. The compressed data is encrypted, adding 5% overhead to the compressed size. Calculate the compressed file size, the encrypted file size, and the total percentage reduction compared to the original file. Explain how the Presentation Layer improves transmission efficiency and security.

Calculation:

Original size = 900 MB

Compressed size = $900 \times (1 - 0.40) = 900 \times 0.60 = 540$ MB

Encryption overhead = 5% of 540 MB = 27 MB

Encrypted (final) size = $540 + 27 = 567$ MB

Net reduction = $900 - 567 = 333$ MB

Percentage reduction = $(333 \div 900) \times 100 = 37\%$

Result:

- Compressed file size = 540 MB
- Encrypted file size = 567 MB
- Total percentage reduction compared to original = 37%

Explanation:

The Presentation Layer is responsible for data representation, translation, compression, and encryption. Compression reduces the number of bits that must be transmitted, which lowers transmission time and bandwidth consumption, directly improving transmission efficiency.

Encryption, applied after compression, converts the data into a secure, unreadable format for anyone without the proper key, protecting confidentiality during transit. Even though encryption adds a small overhead (5% here), the overall data volume sent (567 MB) is still well below the original 900 MB, so the layer achieves both improved efficiency and improved security together.

Question 7

An FTP application transfers 3 GB of data over a network with an effective throughput of 120 Mbps. The Application Layer divides the data into requests of 3 MB each. Calculate the total number of requests generated and estimate the total transmission time in seconds. Discuss how the Application Layer provides reliable services to users.

Calculation:

Total data = 3 GB = 3000 MB

Request size = 3 MB

Number of requests = $3000 \div 3 = 1000$ requests

Total data in bits = 3 GB = 3×10^9 bytes $\times 8 = 24 \times 10^9$ bits

Transmission time = $24,000,000,000 \div 120,000,000 = 200$ seconds

Result:

- Total number of requests generated = 1000
- Total transmission time ≈ 200 seconds

Explanation:

The Application Layer (here, FTP) provides reliable services to users by presenting a simple, well-defined interface (commands and requests) that hides the complexity of the underlying network from the user, while relying on the Transport Layer beneath it for guaranteed, in-order, error-checked delivery of each request.

It also manages user authentication, file/directory operations, and status/error reporting (such as confirming successful transfer of each 3 MB request), so users receive clear feedback and the application itself can retry failed requests, ensuring the overall file transfer completes reliably even though it is broken into many smaller requests.

Question 8

A packet experiences the following processing delays at different OSI layers: Application Layer = 4 ms, Transport Layer = 3 ms, Network Layer = 5 ms, Data Link Layer = 2 ms, and Physical Layer = 1 ms. The propagation delay is 18 ms, and the transmission delay is 12 ms. Calculate the total end-to-end delay experienced by the packet and explain how each layer contributes to the communication process.

Calculation:

Total processing delay = $4 + 3 + 5 + 2 + 1 = 15$ ms

Propagation delay = 18 ms

Transmission delay = 12 ms

Total end-to-end delay = $15 + 18 + 12 = 45$ ms

Result:

- Total end-to-end delay = 45 ms

Explanation:

Each OSI layer adds its own processing delay while performing its specific function: the Application Layer prepares and interprets user data (4 ms), the Transport Layer handles segmentation and reliability checks (3 ms), the Network Layer performs routing and logical addressing decisions (5 ms), the Data Link Layer handles framing and error detection on the link (2 ms), and the Physical Layer converts data into signals for transmission (1 ms).

Beyond these processing delays, transmission delay (12 ms) accounts for the time needed to push all the packet's bits onto the medium, and propagation delay (18 ms) accounts for the time the signal takes to physically travel across the medium to the receiver; together with the cumulative layer-processing delay, these combine to give the total 45 ms end-to-end delay experienced by the packet.

Question 9

A network transfers 80 MB of useful data using frames of 1600 bytes, where each frame contains 80 bytes of protocol overhead added by different OSI layers. Calculate the total number of frames required, the total overhead transmitted, and the transmission efficiency as a percentage. Explain how protocol overhead affects network performance.

Calculation:

Useful data per frame = $1600 - 80 = 1520$ bytes

Total useful data = 80 MB = 80,000,000 bytes

Number of frames = $\lceil 80,000,000 \div 1520 \rceil = 52,632$ frames

Total overhead = $52,632 \times 80 = 4,210,560$ bytes

Total data transmitted = $52,632 \times 1600 = 84,211,200$ bytes

Transmission efficiency = $(80,000,000 \div 84,211,200) \times 100 \approx 95.00\%$

Result:

- Total number of frames required = 52,632
- Total overhead transmitted $\approx 4,210,560$ bytes
- Transmission efficiency $\approx 95.00\%$

Explanation:

Protocol overhead consists of the header and trailer fields (addressing, sequencing, error-checking, control information) that each OSI layer adds to the actual user data. While necessary for reliable, addressed, and error-checked delivery, this overhead consumes part of the available bandwidth without carrying useful payload.

As overhead increases relative to frame size, transmission efficiency decreases and more frames (and more total bytes) are needed to deliver the same amount of useful data, increasing transmission time and reducing effective throughput; network designers therefore try to balance frame size and overhead to keep efficiency high without sacrificing reliability.

Question 10

A hospital transfers a patient's medical record of 240 MB through an IoT-enabled healthcare network. The Presentation Layer compresses the data by 30%, the Transport Layer divides the compressed data into 1400-byte segments, and the Data Link Layer adds 60 bytes of overhead to each frame. Calculate the compressed file size, the total number of segments transmitted, the total protocol overhead introduced, and the final amount of data transmitted across the Physical Layer. Explain how multiple OSI layers work together to ensure secure and reliable communication.

Calculation:

Original size = 240 MB

Compressed size = $240 \times (1 - 0.30) = 240 \times 0.70 = 168 \text{ MB} = 168,000,000 \text{ bytes}$

Number of segments = $168,000,000 \div 1400 = 120,000 \text{ segments}$

Data Link overhead = $120,000 \times 60 = 7,200,000 \text{ bytes}$

Final data across Physical Layer = $168,000,000 + 7,200,000 = 175,200,000 \text{ bytes} \approx 175.2 \text{ MB}$

Result:

- Compressed file size = 168 MB
- Total number of segments transmitted = 120,000
- Total protocol overhead introduced = 7,200,000 bytes ($\approx 7.2 \text{ MB}$)
- Final data transmitted across the Physical Layer $\approx 175.2 \text{ MB}$

Explanation:

Multiple OSI layers cooperate to ensure this sensitive medical record is delivered securely and reliably: the Presentation Layer compresses (and typically encrypts) the record to reduce size and protect patient confidentiality, the Transport Layer segments the compressed data and provides end-to-end reliability through sequencing, acknowledgements, and retransmission of lost segments.

The Network Layer routes each segment's packets across the IoT healthcare network using logical addressing, the Data Link Layer frames each packet with error-detection overhead to catch corruption on each physical hop, and the Physical Layer transmits the resulting bits over the medium — together forming a layered pipeline where each layer's function builds on the one below it to guarantee that the complete, accurate, and confidential medical record reaches its destination.